

# Projeto Aquecedor

Tabela Verdade

| $L_H$ | $L_L$ | $T_H$ | $T_L$ | Entrada | Saída | Aquecedor (H) | Alarme |
|-------|-------|-------|-------|---------|-------|---------------|--------|
| 0     | 0     | 0     | 0     | 1       | 0     | 0             | 0      |
| 0     | 0     | 0     | 1     | 1       | 0     | 1             | 0      |
| 0     | 0     | 1     | 0     | 1       | 0     | 0             | 1      |
| 0     | 0     | 1     | 1     | 0       | 0     | 0             | 1      |
| 0     | 1     | 0     | 0     | 1       | 1     | 0             | 0      |
| 0     | 1     | 0     | 1     | 1       | 0     | 1             | 0      |
| 0     | 1     | 1     | 0     | 1       | 0     | 0             | 1      |
| 0     | 1     | 1     | 1     | 0       | 0     | 0             | 1      |
| 1     | 0     | 0     | 0     | 0       | 0     | 0             | 1      |
| 1     | 0     | 0     | 1     | 0       | 0     | 0             | 1      |
| 1     | 0     | 1     | 0     | 0       | 0     | 0             | 1      |
| 1     | 0     | 1     | 1     | 0       | 0     | 0             | 1      |
| 1     | 1     | 0     | 0     | 0       | 1     | 0             | 0      |
| 1     | 1     | 0     | 1     | 0       | 0     | 1             | 0      |
| 1     | 1     | 1     | 0     | 0       | 0     | 0             | 1      |
| 1     | 1     | 1     | 1     | 0       | 0     | 0             | 1      |

Mapas de K

|                                 | $\overline{T_H} \overline{T_L}$ | $\overline{T_H} T_L$ | $T_H \overline{T_L}$ | $T_H T_L$ | Entrada                                                         |
|---------------------------------|---------------------------------|----------------------|----------------------|-----------|-----------------------------------------------------------------|
| $\overline{L_H} \overline{L_L}$ | 1                               | 1                    | 0                    | 1         |                                                                 |
| $\overline{L_H} L_L$            | 1                               | 1                    | 0                    | 1         | $\overline{L_H} \overline{T_H} + \overline{L_H} \overline{T_L}$ |
| $L_H \overline{L_L}$            | 0                               | 0                    | 0                    | 0         |                                                                 |
| $L_H L_L$                       | 0                               | 0                    | 0                    | 0         |                                                                 |

|                                 | $\overline{T_H} \overline{T_L}$ | $\overline{T_H} T_L$ | $T_H \overline{T_L}$ | $T_H T_L$ | Saída                               |
|---------------------------------|---------------------------------|----------------------|----------------------|-----------|-------------------------------------|
| $\overline{L_H} \overline{L_L}$ | 0                               | 0                    | 0                    | 0         |                                     |
| $L_H \overline{L_L}$            | 1                               | 0                    | 0                    | 0         | $L_L \overline{T_H} \overline{T_L}$ |
| $L_H L_L$                       | 1                               | 0                    | 0                    | 0         |                                     |
| $\overline{L_H} L_L$            | 0                               | 0                    | 0                    | 0         |                                     |

D S T Q Q S S  
D L M M J V S

|                                 | $\overline{F_H} \overline{F_C}$ | $\overline{F_H} F_C$ | $F_H \overline{F_C}$ | $F_H F_C$ | Aquecedor (H)                                                |
|---------------------------------|---------------------------------|----------------------|----------------------|-----------|--------------------------------------------------------------|
| $\overline{L_H} \overline{L_L}$ | 0                               | 1                    | 0                    | 0         |                                                              |
| $\overline{L_H} L_L$            | 0                               | 1                    | 0                    | 0         | $\overline{L_H} \overline{F_H} F_C + L_L \overline{F_H} F_C$ |
| $L_H \overline{L_L}$            | 0                               | 1                    | 0                    | 0         |                                                              |
| $L_H L_L$                       | 0                               | 0                    | 0                    | 0         |                                                              |

|                                 | $\overline{F_H} \overline{F_C}$ | $\overline{F_H} F_C$ | $F_H \overline{F_C}$ | $F_H F_C$ | Alarme                                |
|---------------------------------|---------------------------------|----------------------|----------------------|-----------|---------------------------------------|
| $\overline{L_H} \overline{L_L}$ | 0                               | 0                    | 1                    | 1         |                                       |
| $\overline{L_H} L_L$            | 0                               | 0                    | 1                    | 1         | $L_H \overline{L_L} + \overline{F_H}$ |
| $L_H \overline{L_L}$            | 0                               | 0                    | 1                    | 1         |                                       |
| $L_H L_L$                       | 1                               | 1                    | 1                    | 1         |                                       |